

EEEC1303 DIGITAL LOGIC CIRCUITS

TRUCK OVERLOAD MONITORING AND NOTIFICATION SYSTEM

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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This Project Work (EEB1303 DIGITAL LOGIC CIRCUITS) report has been submitted for the V Semester Project viva voce Examination held on

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EXTERNAL EXAMINER

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PEO2: Graduates will pursue higher studies and succeed in academic/research careers.

PEO3: Graduates will be a successful entrepreneur in creating jobs related to Electrical and Electronics Engineering /allied disciplines.

PEO4: Graduates will practice ethics and have habit of continuous learning for their success in the chosen career.

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- **PO1 Engineering knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals, and an engineering specialization to develop solutions to complex engineering problems.
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- **PO4 Conduct investigations of complex problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

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- **PO11 Life-long learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

The following are the Program Specific Outcomes of Engineering Students:

PSO 01: Power Systems Engineering

Apply comprehensive knowledge of electrical power systems to analyze, design and manage generation, transmission, and distribution networks, integrating emerging technologies and sustainable practices to contribute effectively to industry and societal needs.

PSO 02: Power Electronics and Drives

Design and develop efficient, reliable power electronic systems and motor drives with an emphasis on sustainable energy use, and inclusive technology solutions that address societal and environmental challenges.

PSO 03: Electronics and Instrumentation Engineering

Develop intelligent instrumentation and embedded systems for accurate measurements, monitoring and control in interdisciplinary domains addressing societal needs.

Abstract (key words)	POs Mapping
Overload detection, real-time monitoring, vehicle identification, weight analysis, threshold alert, automated notification, road safety, load management, smart transport system, safety compliance, alert mechanism, transportation monitoring.	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PSO1, PSO2, PSO3

SDG NO	SDG Goal	MAPPING JUSTIFICATION
SDG 9	Industry, Innovation, and Infrastructure	The project supports SDG 9 by using smart technology to prevent truck overloading, protecting road infrastructure and improving transport safety.

ABSTRACT

The Truck Overload Monitoring and Notification System is developed to address the growing problem of overloaded vehicles, which contribute to road deterioration, vehicle breakdowns, and increased accident risks. This system focuses on accurately measuring the load carried by a truck and continuously monitoring it throughout operation. The measured weight is compared against a predefined safety threshold, enabling the system to identify overloading in real time. When an overload condition occurs, the system automatically generates visual and audible warnings for the driver and sends instant notifications to the concerned authorities for immediate action. The model also incorporates vehicle identification, ensuring that load records are correctly mapped to the corresponding truck, improving data reliability and traceability. By providing continuous monitoring, automated alerts, and reliable information flow, this system enhances road safety, supports better regulatory enforcement, and promotes responsible freight management. It offers a cost-effective, efficient, and scalable solution to minimize human error, reduce infrastructure damage, and improve the overall safety of transportation networks.

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LIST OF ABBREVIATIONS

S .No	ABBREVIATION	EXPANSION
1	GSM	Global System for Mobile Communication
2	GPS	Global Positioning System
3	RFID	Radio Frequency Identification
4	UART	Universal Asynchronous Receiver and Transmitter
5	UID	Unique Identification Number
6	LCD	Liquid Crystal Display
7	SMS	Short Message Service
8	IDE	Integrated Development Environment

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Overloaded trucks accelerate road wear and raise the probability of accidents, which has a substantial impact on road safety and transportation infrastructure. Weighbridges and physical inspections frequently monitor truckloads, but these methods are often labor-intensive, slow, and prone to human errors. Developments in sensor technology and embedded systems enable the implementation of automated solutions for real-time load monitoring. These technologies are used by the Truck Overload Detection with Notification System to continuously monitor truck weights and promptly notify authorities if restrictions are exceeded. In addition, modern embedded platforms allow integration with GSM based communication modules, ensuring immediate alerts even from remote locations. Automated monitoring not only reduces manual effort but also enhances transparency by providing accurate and tamper-proof data. This method seeks to increase overall road safety, save maintenance costs, and improve regulatory compliance by combining automated detection with timely, technology-driven alerts.

1.2 PROBLEM STATEMENT

Overloading of trucks is a persistent problem in the transportation sector, leading to serious consequences such as road surface damage, reduced fuel efficiency, increased vehicle wear and tear, and a higher risk of accidents. Manual checking of truck loads at checkpoints is time-consuming, unreliable, and often ineffective due to human error. In many cases, overloaded vehicles bypass weighbridges or manipulate measurements, further worsening the issue. Currently, there is no efficient system to automatically detect in real time

when a truck exceeds its permissible load limit or to notify authorities immediately. This lack of continuous monitoring results in delayed action and allows violations to go unnoticed .By integrating sensors, microcontrollers, and wireless communication technologies, such a system can significantly improve monitoring accuracy, reduce manual workload, and enhance regulatory compliance. Additionally, real-time notifications enable authorities to take immediate corrective action, thereby preventing accidents, reducing road damage, and promoting safer transportation operations.

1.3 OBJECTIVE

The main objective of the Truck Overload Detection and Notification System is to develop an automated solution that accurately measures the load carried by a truck and identifies any overload condition in real time. The system aims to ensure safer transportation by continuously monitoring load weight, comparing it with a predefined threshold, and alerting the driver and concerned authorities when the permissible limit is exceeded. Additional objectives include reducing manual inspection efforts, preventing road and vehicle damage caused by overloading.

1.4 SCOPE OF THE PROJECT

This project focuses on developing an automated system to detect overloaded trucks and send real-time alerts to authorities. The system identifies vehicles using RFID, measures load through sensors, and uses a GSM module to send notifications when the permissible limit is exceeded. It aims to reduce manual inspection, minimize human errors, and ensure quick action through instant alerts. The scope is limited to overload detection and SMS notification within a specific monitoring area, without features like GPS tracking or cloud storage, which can be added in future improvements.

1.5 LITERATURE REVIEW

Paper 1 : RFID-Based Vehicle Positioning and Its Applications in Connected Vehicles

Authors: Jianqiang Wang, Daiheng Ni & Keqiang Li

Year: 2023

Description: This paper proposes a method for positioning vehicles using RFID tags: when a vehicle passes over an RFID tag embedded (or placed) at known locations, the vehicle's position is determined precisely from the tag's stored coordinates. The approach was intended to support connected-vehicle applications, particularly in scenarios where GPS might be unreliable (for example, "urban canyons" or tunnels). For your overload-detection system, this kind of RFID-based positioning/identification could be adapted to reliably identify vehicles (via RFID) as they pass weigh-in or scanning points ensuring that load measurements (from load cells) are correctly associated with the right vehicle identity.

Paper 2 : An Architecture of the Vehicle Tracking System Based on RFID

Authors: Jian Xin Deng

Year: 2022

Description: This paper describes a multi-layered architecture for a vehicle tracking system using RFID technology. The architecture integrates databases, RFID tags, readers, data centers, networks and user interfaces to enable automatic tracking of vehicles. When RFID readers detect a vehicle's tag, the system logs the event, linking the vehicle's UID with positional data stored in the system database. In a load-monitoring context, such an architecture can be useful to ensure that each weight measurement (from load-cells) is recorded along with the correct vehicle identity (via RFID), facilitating management,

record-keeping, and preventing identity mismatches or misuse.

Paper 3 : The Analysis and Design of Weighbridge Network Monitoring System Based on RFID

Authors: Lv Qing Yang, Hao Wang

Year: 2020

Description: This paper describes the design and implementation of a weighbridge-network monitoring system that integrates RFID, infrared locators, and anti-fraud monitoring. The authors present both the physical architecture (hardware) and software architecture (three-tier software design), along with the interface design for RFID and other sensors. Through test cases and validation, they show that the system works correctly and can reliably monitor vehicle weights while linking the weight data with vehicle identity via RFID.

Paper 4 : Automatic Weight Monitoring Using RFID and Load Cell

Authors: M. Lakshmi & K. Hariprasad

Year: 2019

Description: In this work, the authors propose a system for toll-gate / weighing-station applications that uses RFID for vehicle identification, a load cell for weight measurement, and a microcontroller to manage the system. When a vehicle arrives, its RFID tag is read to identify it; then the load cell measures the load/weight. If the weight exceeds a threshold (i.e. overload condition), the system captures vehicle details (e.g. via image), and triggers alerts or logging. This scheme matches closely with your overload-detection project: combining identity verification (RFID), weight measurement (load cell), and decision logic (overload alert).

CHAPTER 2

PROPOSED METHODOLOGY

2.1 PROPOSED WORK

The proposed system continuously measures the truck's load using a load cell and HX711 amplifier, and the data is processed by an Arduino Uno. The measured weight is compared with a predefined threshold to detect any overload condition. If the load is within the limit, the system displays the weight on the LCD. When the load exceeds the permissible limit, the system shows an overload warning and automatically sends a notification to a predefined mobile number through the GSM module. The system operates in real time, monitors any change in load, and provides a low-cost and reliable solution for preventing overloading and improving road safety.

2.2 BLOCK DIAGRAM

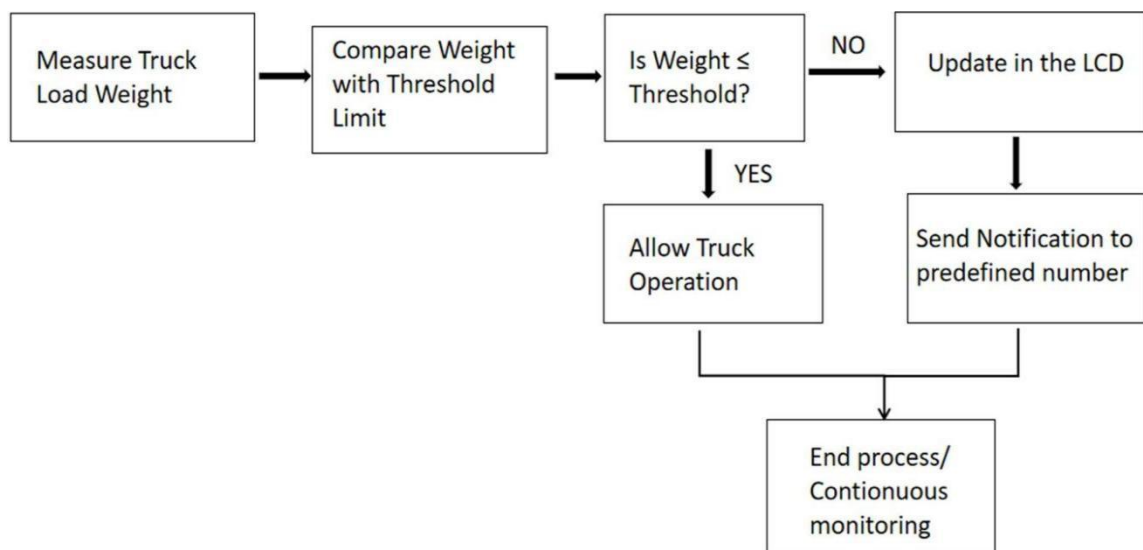


Fig 2.1 Work Flow

2.3 HARDWARE IMPLEMENTATION

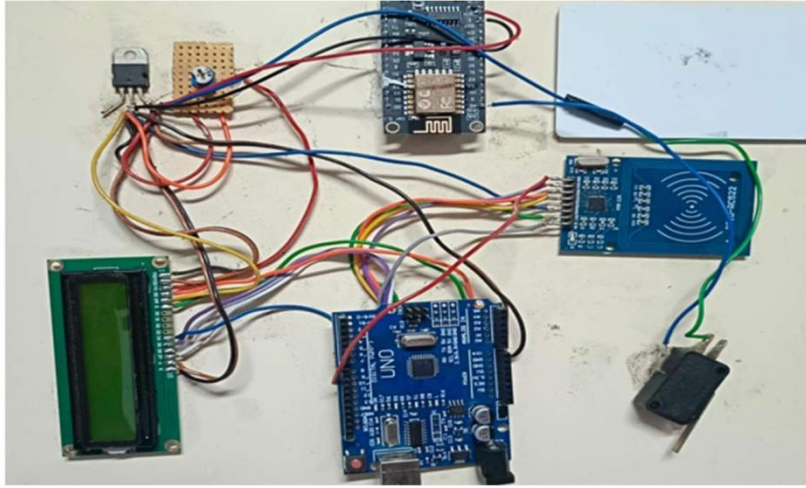


Fig 2.2 Hardware Setup

2.4 HARDWARE DESCRIPTION

The hardware setup for the Truck Overload Monitoring and Notification System integrates sensing, processing, display, and communication modules into a single working prototype. The core control is handled by an Arduino Uno, which receives data from the connected sensors and modules. A load cell sends weight signals to the microcontroller through an HX711 amplifier. An LCD display is used to show the measured weight and system status. For communication and alert functions, the system includes an ESP8266 Wi-Fi module, which enables data transmission or notification features. An RFID module is included for vehicle identification or access verification, and a micro limit switch is used to detect physical activation or weight platform movement. A voltage regulator and mini power board ensure stable power supply to all modules.

2.5 COMPONENTS DESCRIPTION

Microcontroller (Arduino / ESP32)



Fig 2.3 Microcontroller

The Arduino Uno acts as the main controller of the system, processing input signals from sensors and modules. It reads data from the RFID reader, limit switch, and display, then performs logic operations. It ensures smooth communication between components and controls the workflow of the entire project.

LCD Display (16x2 / 20x4)



Fig 2.4 LCD display

The LCD display shows the measured weight, RFID status, and overload warnings to the user. It provides real-time visual feedback, making the system easy to understand and operate. The display enhances user interaction by showing the system's current state.

RFID module



Fig 2.5 RFID module

The RC522 RFID reader identifies the vehicle or user by reading the unique ID stored in RFID tags. It works at short range and ensures only authorized vehicles are detected by the system. This helps automate identification without manual entry, increasing accuracy and security.

Micro Limit Switch



Fig 2.7 Micro Limit Switch

The limit switch acts as a position-detection sensor, indicating when the truck or miniature model is properly placed. When pressed, it signals the controller to begin weight or process measurement. This prevents incorrect readings and ensures the system starts only at the right moment.

CHAPTER 3

SYSTEM FRAMEWORK

3.1 Software Required

Arduino IDE

Used for writing, compiling, and uploading the embedded program to the Arduino microcontroller. It provides an easy-to-use interface and supports the required libraries for RFID, GSM, and LCD modules.

Arduino Libraries

SPI Library – Enables communication between the Arduino and RFID reader (MFRC522).

MFRC522 Library – Used to read and process RFID tag information.

SoftwareSerial Library – Allows the GSM module (SIM900) to communicate through digital pins.

Liquid Crystal Library – Controls the LCD display for showing system messages.

Serial Monitor / Serial Plotter

Built into the Arduino IDE, used for debugging, monitoring the RFID data, and checking SMS sending responses.

3.2 ALGORITHMS

System Initialization

The system begins by setting up all hardware components required for proper operation. The Arduino initializes the SPI communication protocol to communicate with the MFRC522 RFID reader. The GSM module (SIM900) is configured to operate in text mode for sending SMS alerts. The LCD display is also initialized to show messages to the user. Once all modules are successfully

initialized, the LCD displays a welcome message followed by the prompt “Swipe Your Card,” indicating the system is ready to scan RFID tags.

RFID Card Detection

The RFID reader continuously monitors its magnetic field to detect nearby cards. The system uses the function `PICC_IsNewCardPresent()` to check if a new card has entered the scanning range. When a card is detected, the reader captures its UID using `PICC_ReadCardSerial()`. The UID consists of 4 bytes that uniquely identify each card. The system also verifies the type of RFID card to ensure it is a supported MIFARE Classic card. Only after this validation does the system proceed to the next stage, preventing errors from unsupported or unreadable cards.

UID Verification and Authentication

Once the card’s UID is read, the system compares it against a list of predefined UIDs stored in the program. Each UID corresponds to a registered vehicle or user. This authentication step ensures that only authorized cards trigger system actions. If the scanned UID matches one of the stored UIDs, the card is accepted, and the system identifies the specific user. If the UID does not match any stored values, the system classifies the user as “Unregistered” and ignores the card. This verification step provides security and prevents unauthorized access or false alerts.

Action Execution (SMS Sending)

After a successful UID match, the system performs the assigned function linked to that card. For example, if the card belongs to a vehicle with overload, the system sends an overload fine alert to the registered phonenumber. If it is configured for attendance, the system sends an attendance confirmation SMS.

The GSM module sends messages using AT commands such as AT+CMGF=1 for text mode and AT+CMGS to specify the recipient number. The LCD simultaneously displays real-time feedback such as “Scanning Vehicle” or “Message Sent.” This ensures the user is informed about the process while the system sends accurate messages automatically.

System Reset and Looping

After completing the action, the system resets the display and returns to its standby mode. The LCD clears previous messages and again shows “Swipe Your Card.” This marks the end of one complete cycle of detection, verification, and notification. The loop function then restarts the scanning process, enabling the system to operate continuously without the need for manual resetting. This repeated looping ensures the system is always ready for the next card, making it suitable for real-time monitoring applications such as vehicle load management or attendance tracking.

3.3 SYSTEM ARCHITECTURE

System Design

The system architecture outlines the overall structure, functionality, and interaction between different components that make up the Truck Overload Detection and Notification System. It provides a clear view of how the system captures vehicle identity, measures load, processes data, and communicates alerts. This architecture ensures that the system operates in a coordinated, reliable, and real-time manner. It combines hardware modules such as sensors, RFID readers, GSM modules, and microcontrollers with software algorithms that manage detection, verification, and message automation. By structuring the system into well-defined blocks, the architecture ensures modularity, ease of troubleshooting, and scalability for future enhancements.

Hardware Architecture

The hardware architecture consists of interconnected electronic components, each serving a specific functional purpose. The load cell acts as the primary sensing device and detects changes in the truck's weight. The Arduino or microcontroller functions as the central processing unit, collecting sensor data and interpreting it to determine overload conditions. The RFID module identifies each vehicle uniquely through RFID cards or tags. The LCD module provides on-site visual feedback indicating scanning status, overload detection, or message transmission. Proper wiring, signal conditioning, and power supply stability are essential to ensure reliable operation of the entire hardware setup.

Software Architecture

The software architecture defines the flow of instructions and logic that control the operation of the hardware components. It starts with initializing all modules such as RFID, GSM, LCD, and sensor libraries. The code continuously monitors the RFID reader for card presence and reads the UID when detected. Once verified, the system checks the weight measured by the load sensor and compares it with the permissible threshold coded in the program. If the loads exceeds the limit, the software executes the GSM alert subroutine to notify concerned personnel. Error handling routines, loop structures for continuous monitoring, and library functions for communication protocols form the foundation of the software architecture.

Functional Modules Description

The system is divided into multiple functional modules, each responsible for a specific task:

Identification Module: Uses RFID technology to uniquely identify each truck. This prevents unauthorized or unknown vehicles from bypassing the system.

Load Sensing Module: Utilizes a load cell or weight sensor to measure the truck load in real time and converts physical weight into an electrical signal.

Processing and Decision Module: The microcontroller processes UID and weight readings, compares values with preset limits, and decides whether the truck is overloaded or not.

Alert/Communication Module: Sends SMS alerts using the GSM module whenever an overload is detected, ensuring immediate communication with authorities.

Display Module: Shows status messages such as "Swipe Card," "Vehicle Identified," "Overload Detected," or "Message Sent" using the LCD display. This modular approach ensures smooth coordination and simplifies debugging and enhancement.

System Integration

System integration ensures seamless communication and coordination among hardware and software elements. The RFID module communicates with the microcontroller using the SPI protocol, enabling fast data exchange. The GSM module uses UART serial communication to send AT commands for SMS transmission. Load cell data is received through an analog input or amplifier circuit for accurate readings. Power management is also integrated to ensure that all components receive stable voltage, which is critical for accurate sensing and communication. This integrated architecture enables the system to operate reliably and continuously in real-world environments.

CHAPTER 4

TESTING AND VALIDATION

4.1 Performance and Stability Analysis

Test Objectives

The primary objective of testing is to ensure that the system accurately detects overloaded trucks and performs reliably under various operating conditions. Testing aims to verify the proper functioning of sensor readings, RFID identification, and GSM-based alert notifications. It also helps confirm that the algorithm responds correctly to real-time load variations. Overall, the goal is to validate that the system meets the requirements of safety, accuracy, and efficiency.

Component-Level Testing

Each component of the system was tested individually before integration to ensure proper operation. The load cell was calibrated and checked for precise weight readings, while the RFID reader was tested for correct UID detection. The GSM module was validated by sending multiple test messages to verify network reliability. Testing each component individually helped identify and fix errors early in the development stages.

Functional Testing

Functional testing evaluated how the system behaves when processing different scenarios such as normal loads, overloaded conditions, and scanning unregistered cards. The system correctly identified registered vehicles, measured the load, and triggered alerts when limits were exceeded. It also demonstrated stable operation during repeated scanning and weight changes. This testing confirmed that all system functions work as intended.

Performance and Reliability Testing

Performance testing focused on response time, SMS delivery speed, and accuracy of load measurements. The system consistently sent alerts within a few seconds of detecting overload conditions. Reliability tests involved running the system continuously over extended periods to ensure stable operation. The results showed that the system maintains accuracy and communication consistency without failures.

Test Results

The overall testing outcomes confirmed that the system is accurate, responsive, and reliable for real-time overload detection. All hardware components performed well, and software logic executed without errors. The GSM alerts were delivered promptly, and RFID identification worked smoothly. Based on the results, the system is ready for practical deployment with strong performance in all key areas.

4.2 RESULT AND DISSCUSSION

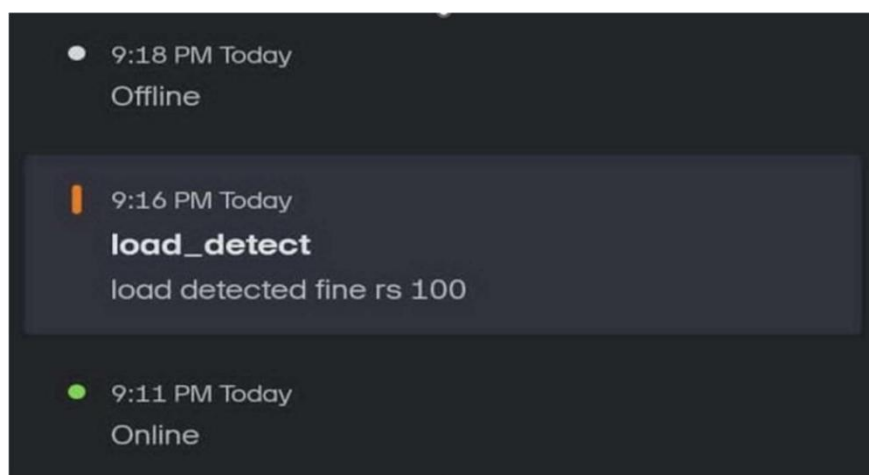


Fig 4.1 Result

The Truck Overload Monitoring and Notification System was successfully implemented and tested, demonstrating reliable performance in real-time weight monitoring and overload detection. The load cell and HX711 amplifier provided accurate and stable weight measurements, which were continuously displayed on the LCD. When the load exceeded the preset threshold, the system instantly identified the overload and triggered an alert message on the LCD while simultaneously sending an SMS notification to the authorized mobile number through the GSM module. The RFID module accurately identified the truck or driver using unique tags, ensuring proper authentication and record maintenance. Overall, the project demonstrates that a low-cost microcontroller-based design can deliver accurate overload detection and efficient alert mechanisms suitable for real-world transportation application

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

A smart overload monitoring and notification system is developed to address the issue of trucks carrying excessive weight, which contributes to road damage, mechanical failures, and increased accident risk. The system continuously measures the vehicle's load and compares it with a predefined threshold to detect any overload condition in real time. When the weight exceeds the permitted limit, it triggers immediate warnings for the driver and automatically sends notifications to the relevant authorities for timely intervention. The model also integrates vehicle identification to ensure accurate load tracking and reliable record management. By providing real-time monitoring, automated alerts, and improved accuracy, the system enhances transport safety, supports better enforcement of regulations, and encourages responsible load management. Overall, it offers an efficient and cost-effective solution to reduce human error, protect infrastructure, and promote safer transportation practices.

The system can be further improved by integrating cloud-based data storage to maintain long-term records of vehicle load history. GPS-based route tracking can be added to monitor the truck's movement and identify areas where overloading frequently occurs. Advanced data analytics and dashboards may help authorities make informed decisions and strengthen transport regulations. The system can also be upgraded to support mobile app monitoring, automatic report generation, and integration with government transport databases. In the future, stronger and more accurate sensors can be used to support heavy-duty commercial vehicles, making the system more reliable for real-world implementation.

CHAPTER 6

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APPENDIX

```
/* terminal1 program
```

```
-----
```

```
* Pin layout should be as follows:
```

```
* Signal   Pin           Pin           Pin
*          Arduino Uno   Arduino Mega   MFRC522 board
```

```
* -----
```

```
* Reset    9             5             RST
* SPI SS    10           53             SDA
* SPI MOSI  11           51             MOSI
* SPI MISO  12           50             MISO
* SPI SCK   13           52             SCK
```

```
* voltage 3.3v
```

```
*/
```

```
// gsm module is connected with pin number 7 and pin number 8 of the arduino.
```

```
#include <SPI.h>
```

```
#include <MFRC522.h>
```

```
#include <LiquidCrystal.h>
```

```
#include <SoftwareSerial.h>
```

```
SoftwareSerial SIM900(7, 8); // gsm module connected here
```

```
String textForSMS;
```

```
#define SS_PIN 10
```

```
#define RST_PIN 9
```

```
MFRC522 mfrc522(SS_PIN, RST_PIN); // Create MFRC522 instance.
```

```

// lcd pins
#define rs A1
#define en A2
#define d4 6
#define d5 5
#define d6 4
#define d7 3

// initialize the library with the numbers of the interface pins
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);

// parents numbers

String f1001 = "+918220704543"; // student1 father cell phone number
String f1002 = "+918220704543";
String f1003 = "+918220704543";

void setup() {
    Serial.begin(9600);    // Nodemcu is connected over here
    randomSeed(analogRead(0));
    SIM900.begin(9600); // original 19200. while enter 9600 for sim900A
    SPI.begin();          // Init SPI bus
    mfrc522.PCD_Init();    // Init MFRC522 card
    //Serial.println("Scan a MIFARE Classic PICC to demonstrate Value
    Blocks.");

    // set up the LCD's number of columns and rows:
    lcd.begin(16, 2);
    // Print a message to the LCD.

```



```

lcd.print("rfid based load moniter");
delay(3000);
lcd.clear();
lcd.print("Swip Your Card");
delay(1000);
}
void loop() {
    // Prepare key - all keys are set to FFFFFFFFFFh at chip delivery from
the factory.
    MFRC522::MIFARE_Key key;
    for (byte i = 0; i < 6; i++) {
        key.keyByte[i] = 0xFF;
    }
    // Look for new cards
    if ( ! mfrc522.PICC_IsNewCardPresent()) {
        return;
    }
    // Select one of the cards
    if ( ! mfrc522.PICC_ReadCardSerial()) {
        return;
    }
    // Now a card is selected. The UID and SAK is in mfrc522.uid.

    // Dump UID
    Serial.print("Card UID:");
    for (byte i = 0; i < mfrc522.uid.size; i++) {
        // Serial.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");
        // Serial.print(mfrc522.uid.uidByte[i], DEC);
    }
}

```

```

    }
    Serial.println();

    // Dump PICC type
    byte piccType = mfrc522.PICC_GetType(mfrc522.uid.sak);
    // Serial.print("PICC type: ");
    //Serial.println(mfrc522.PICC_GetTypeName(piccType));
    if (    piccType != MFRC522::PICC_TYPE_MIFARE_MINI
        &&    piccType != MFRC522::PICC_TYPE_MIFARE_1K
        &&    piccType != MFRC522::PICC_TYPE_MIFARE_4K) {
        //Serial.println("This sample only works with MIFARE Classic
cards.");
        return;
    }
    // defining Cards here

    // student1
    if( (mfrc522.uid.uidByte[0] == 57) && (mfrc522.uid.uidByte[1] == 240)
    && (mfrc522.uid.uidByte[2] == 01) && (mfrc522.uid.uidByte[3] == 04) ) //
student1 card
    {
        lcd.clear();
        lcd.print("scanning vehicle");
        delay(500);
        Serial.println("student 1");

        // for gsm
        sendsms("your vehicle over load fine RS100", f1001);

```

```

    delay(1000);

    lcd.setCursor(0, 1);
    lcd.print("message sent");
    delay(2000);

    lcd.clear();
    lcd.print("Swip Your Card");
    delay(1000);
}

        // student2
        if( (mfrc522.uid.uidByte[0] == 26) && (mfrc522.uid.uidByte[1] == 141)
        && (mfrc522.uid.uidByte[2] == 07) && (mfrc522.uid.uidByte[3] == 177) ) //
student2 card
    {
        lcd.clear();
        lcd.print("AttendanceMarked");
        delay(500);
        Serial.println("student 2");

        // for gsm
        sendsms("pavithra/BCA/3YR is Present", f1002);
        delay(1000);

        lcd.setCursor(0, 1);
        lcd.print("message sent");
        delay(2000);

```

```

        lcd.clear();
        lcd.print("Swip Your Card");
        delay(1000);
    }
    else
        Serial.println("unregistered user");
}

void sendsms(String message, String number)
{
    String mnumber = "AT + CMGS = \""+number+"\"";
    SIM900.print("AT+CMGF=1\r");
    delay(1000);
    SIM900.println(mnumber); // recipient's mobile number, in international
    format

    delay(1000);
    SIM900.println(message); // message to send
    delay(1000);
    SIM900.println((char)26); // End AT command with a ^Z,
    ASCII code 26
    delay(1000);
    SIM900.println();
    delay(100); // give module time to send SMS
    // SIM900power();
}

```